

Prompt 1: Initial Code Generation

Write a beginner-friendly C++ console program that generates funny roast messages.
The program should ask the user to enter their name.
Store at least 10 roast templates using a {name} placeholder.
Randomly select one roast each time the program runs.
Replace {name} with the actual user input.
Handle empty input properly and display an error message.
Add comments in the code so that beginners can understand it easily.

◇ Prompt 2: Code Improvement / Refinement

Improve the previous C++ roast generator program.
Make sure that if {name} appears multiple times in a roast template, it is replaced everywhere.
Also make the code cleaner, readable, and beginner-friendly.

◇ Prompt 3: Bug Identification

My C++ roast generator program is not replacing {name} correctly in all cases.
Sometimes only one occurrence is replaced.
Why is this happening?

◇ Prompt 4: Bug Fix Request

Fix the issue in my C++ program where {name} is not replaced everywhere in the string.
Modify the code so that all occurrences of {name} are replaced correctly.
Explain the fix in simple terms.

◆ Prompt 1 (top)

Write a beginner-friendly C++ console program that generates funny roast messages. The program should ask the user to enter their name. Store at least 10 roast templates using a {name} placeholder. Randomly select one roast each time the program runs. Replace {name} with the actual user input. Handle empty input properly and display an error message. Add comments in the code so that beginners can understand it easily.

◆ AI Response (Code below prompt)

```
C++

#include <iostream>
#include <vector>
#include <cstdlib> // For rand() and srand()
#include <ctime>   // For time()
using namespace std;

// Function to replace {name} with actual user input
string personalize(string roast, string name) {
    size_t pos = roast.find("{name}");

    if (pos != string::npos) {
        roast.replace(pos, 6, name); // Replace only 1st occurrence
    }
}
```

+ Ask anything

```
C++

}

return roast;
}

int main() {
    string name;

    // Ask user for input
    cout << "Enter your name: ";
    getline(cin, name);

    // Handle empty input
    if (name.empty()) {
        cout << "Error: Name cannot be empty!" << endl;
        return 0;
    }

    // Store roast templates
    vector<string> roasts = {
        "{name}'s code runs so slowly that even dial-up internet feels fast.",
        "If laziness had a brand ambassador, it would be {name}.",
        "{name} doesn't debug code - they negotiate with bugs.",
        "{name}'s coding speed makes glaciers look fast.",
        "Even a calculator gets confused by {name}'s logic.",
        "{name} writes code like it's a suspense thriller - nothing works till the end.",
        "If errors were money, {name} would be a billionaire.",
        "{name}'s keyboard deserves an award for suffering.",
        "{name} writes code so bad, even AI asks for help.",
    };
}
```

+ Ask anything

```

C++
// ...
    "{name} writes code so bad, even AI asks for help.",
    "{name}'s bugs have bugs."
};

// Initialize random seed
srand(time(0));

// Generate random index
int index = rand() % roasts.size();

// Personalize roast
string finalRoast = personalize(roasts[index], name);

// Display output
cout << finalRoast << endl;

return 0;
}

```

OUTOUT

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\Wikita\OneDrive\Desktop\focp2-lab manual> cd "c:\Users\Wikita\OneDrive\Desktop\focp2-lab manual\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter your name: HARSHIT
If errors were money, HARSHIT would be a billionaire.
PS C:\Users\Wikita\OneDrive\Desktop\focp2-lab manual>

```

◆ Prompt 2 (top)

Improve the previous C++ roast generator program.
Make sure that if {name} appears multiple times in a roast template, it is replaced everywhere.
Also make the code cleaner, readable, and beginner-friendly.

◆ AI Response (Improved Code)

```
</> C++  
  
#include <iostream>  
#include <vector>  
#include <cstdlib>  
#include <ctime>  
using namespace std;  
  
// Function to replace ALL occurrences of {name}  
string personalize(string roast, const string& name) {  
    size_t pos = 0;  
  
    // Loop to replace every occurrence of "{name}"  
    while ((pos = roast.find("{name}", pos)) != string::npos) {  
        roast.replace(pos, 6, name);  
        pos += name.length(); // Move forward after replacement  
    }  
  
    return roast;  
}
```

```
</> C++  
  
}  
  
int main() {  
    string name;  
  
    // Ask user for input  
    cout << "Enter your name: ";  
    getline(cin, name);  
  
    // Handle empty input  
    if (name.empty()) {  
        cout << "Error: Name cannot be empty!" << endl;  
        return 0;  
    }  
  
    // Roast templates (some contain multiple {name})  
    vector<string> roasts = {  
        "{name}, your code is so slow that even {name}'s shadow arrives earlier.",  
        "If laziness had a face, it would look exactly like {name}.",  
        "{name} doesn't debug code - {name} just hopes the bugs fix themselves.",  
        "{name}'s coding speed makes glaciers look fast.",  
        "Even a calculator gets confused by {name}'s logic.",  
        "{name} writes code like a mystery - even {name} doesn't know what it does.",  
        "If errors were money, {name} would be a billionaire.",  
        "{name}'s keyboard deserves an award for suffering.",  
        "{name} writes code so bad, even AI asks {name} for help.",  
        "{name}'s bugs have bugs - and they blame {name}."};  
};
```

```
C++  
};  
  
// Initialize random seed for different output each run  
srand(time(0));  
  
// Select random roast  
int index = rand() % roasts.size();  
  
// Personalize selected roast  
string finalRoast = personalize(roasts[index], name);  
  
// Display result  
cout << finalRoast << endl;  
  
return 0;  
}
```

OUTPUT

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
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Enter your name: HARSHIT  
HARSHIT doesn't debug code rçð HARSHIT just hopes the bugs fix themselves.  
PS C:\Users\Wikita\OneDrive\Desktop\focp2-lab manual>
```

Prompt 3: Bug Identification

Prompt:

My C++ roast generator program is not replacing {name} correctly in all cases.
Sometimes only one occurrence is replaced.
Why is this happening?



AI Response:

This issue happens because your program is using the `find()` function only once.

The `find("{name}")` function returns the position of the first occurrence of "{name}" in the string.

After replacing it, your code does not continue searching for other occurrences.

So if "{name}" appears multiple times in a roast template, only the first one gets replaced.

The remaining occurrences stay unchanged.

To fix this, you should use a loop (such as a while loop).

The loop will repeatedly search for "{name}" and rep.  it until no more occurrences are found.



Prompt 4: Bug Fix Request

Prompt:

Fix the issue in my C++ program where {name} is not replaced everywhere in the string.
Modify the code so that all occurrences of {name} are replaced correctly.
Explain the fix in simple terms.



AI Response:

```
⌞ C++  
  
// Fixed function to replace ALL occurrences of {name}  
string personalize(string roast, const string& name) {  
    size_t pos = 0;  
  
    // Keep replacing until no {name} is left  
    while ((pos = roast.find("{name}", pos)) != string::npos) {  
        roast.replace(pos, 6, name);  
        pos += name.length(); // Move forward after replacement  
    }  
  
    return roast;  
}
```



AFTER FIXING ALL BUGS

CODE

```
#include <iostream>

#include <vector>

#include <cstdlib>

#include <ctime>


using namespace std;


// Function to replace {name} with actual user input
string personalize(string roast, string name) {
    size_t pos = roast.find("{name}");
    if (pos != string::npos) {
        roast.replace(pos, 6, name);
    }
    return roast;
}


int main() {
    string name;


    cout << "Enter your name: ";
    getline(cin, name);


    // Handle empty input
    if (name.empty()) {
        cout << "Please enter a valid name!" << endl;
        return 0;
    }
```

```

// Roast templates
vector<string> roasts = {
    "{name} writes code so slow that even a turtle switched to Python.",
    "If procrastination were an Olympic sport, {name} would win gold.",
    "{name}'s debugging style is staring at the screen until the bug leaves.",
    "{name}'s code runs so slowly that dial-up internet feels fast.",
    "If laziness had a brand ambassador, it would be {name}.",
    "{name} doesn't debug code — they negotiate with bugs.",
    "{name}'s code has more errors than a broken calculator.",
    "{name} types so slow, even autocorrect gave up.",
    "{name}'s logic is like WiFi — weak and unstable.",
    "Even AI refuses to fix {name}'s code."
};

// Seed random generator
srand(time(0));

int index = rand() % roasts.size();

string finalRoast = personalize(roasts[index], name);

cout << "\n 🍷 Roast Generator 🍷 " << endl;
cout << finalRoast << endl;

return 0;
}

```

OUTPUT


```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

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Enter your name: harshit

≡f0N Roast Generator ≡f0N
harshit's code has more errors than a broken calculator.
PS C:\Users\Wikita\OneDrive\Desktop\focp2-lab manual> 
```

